

COSIMA Cookbook: a community-driven resource for ocean and sea-ice modelling science

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Software

- [Review](#)
- [Repository](#)
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Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks (Kluyver et al., 2016). It has been developed by the Consortium for Ocean–Sea Ice Modelling in Australia (COSIMA; <https://cosima.org.au>) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean–sea ice model configurations (e.g. Kiss et al., 2020). It is developed openly in a GitHub repository and it combines introductory tutorials with worked analysis examples as Python Jupyter notebooks. All tutorials and examples are exposed through a browsable documentation site built with Sphinx, a documentation generator for Python projects (Komiya et al., 2025) (Figure 1). Tagged versions of the repository are uploaded to a citable archived release (Constantinou et al., 2026).

COSIMA Cookbook

Search Go

Navigation

Lessons

Cooking Tutorials

- Basics
- Advanced

Recipes

Easy Recipes

Advanced Recipes

Regional Specialties

Resources

Contributing

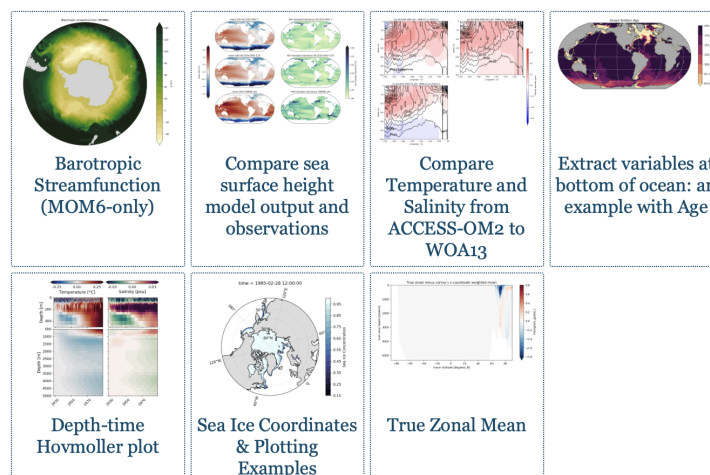
- Getting started with git
- Working on your recipe
- Tips for making a good recipe
- Submitting a Pull Request
- Reviewing existing Pull Requests

Notebook Guidelines

- Thumbnails
- Images in notebooks

GitHub Repository

Easy Recipes



latest

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Figure 1: The COSIMA Cookbook documentation website, showing the browsable Sphinx-based landing page used to navigate tutorials and recipes. The documentation website lives at <https://cosima-recipes.readthedocs.io>.

The COSIMA Cookbook is deliberately organized using a gastronomy theme to help users quickly understand and navigate the website. “Cooking Tutorials” refers to tutorials that teach generic, transferable techniques for handling ocean–sea ice model output. Then comes the main part of any cookbook – its recipes. Here, by “recipes” we mean self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. First come the “Easy Recipes” that provide a good starting point after the tutorials; then the “Advanced Recipes” that are more elaborate and demonstrate advanced analysis techniques, and lastly the “Regional Specialties” contains recipes for regional model configurations (Barnes et al., 2024). Together these sections present the documentation as a browsable collection of lessons and recipes gathered into a single cookbook. The COSIMA Cookbook is one of several examples of using a gastronomy theme for computational workflows, for example Project Pythia Cookbooks for the general geoscience community and the FAIR Cookbook for working with life science data.

Pedagogy and instructional structure are central to the project. The recipes provide the starting point for more elaborate analyses. “Cooking Tutorials” introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused recipes giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions.

Within this structure, the tutorials leverage and demonstrate open source tools for scientific analysis of large data. This includes examples of loading model output through Intake-based

catalogues (Intake developers, 2026), using Xarray (Hoyer & Hamman, 2017) and Dask (Dask Development Team, 2016; Rocklin, 2015) to process very large multi-dimensional datasets in parallel. The recipe collection includes a Lagrangian particle-tracking workflow built with Parcels (Delandmeter & Seville, 2019), as well as analyses that use xgcm (Abernathy et al., 2022) to handle variables defined on staggered finite-volume grids, and xESMF (Zhuang et al., 2025) for regridding model output. Figures are generated using matplotlib (Hunter, 2007) incorporating Cartopy (Elson et al., 2024) for map-based visualisation.

The Cookbook grew from a practical need inside the COSIMA community: the tools used to analyse modern ocean model output are powerful, but the gap between package-level documentation and reproducible end-to-end workflows remains large. Users need to understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate high-dimensional and often very large model output, shared high-performance computing environments, and domain-specific analysis conventions. The Cookbook addresses this gap by packaging reusable workflows in the same medium in which users actually work.

Statement of Need

Analysing or configuring climate models has a steep entry cost. New users must learn how to find datasets, load them efficiently, interpret metadata, operate on multi-dimensional arrays, and produce scientifically meaningful diagnostics. General-purpose libraries such as Xarray (Hoyer & Hamman, 2017) provide the foundations for these tasks, but they do not by themselves show a learner how to translate a research question into a robust analysis workflow for a specific model and computing environment.

That gap is not only pedagogical; it directly affects how quickly model output can be turned into science. For a beginner, becoming productive with a model often requires substantial time just to understand the structure of the output, the relevant conventions, and the computational environment in which analysis is expected to run. There is also substantial scientific labour embedded in many common diagnostics. Some calculations, for example computing cross-contour transports, require non-trivial development, understanding of model numerics, and validation before they can be used confidently. If every new user or project had to reconstruct those workflows independently, a large amount of effort would be repeatedly spent on rebuilding analysis code rather than conducting research using the model output.

The COSIMA Cookbook facilitates knowledge sharing and accelerates research with a domain-specific, openly maintained collection of computational lessons and examples. Its contribution is not a new analysis library; rather, it is a reusable scientific and educational resource that exposes complete, well-documented workflows in the same notebook format in which many researchers actually explore data and communicate methods. By choosing notebooks instead of packaging these examples only as a software library, the project makes intermediate reasoning, methodological choices, and practical execution details visible to learners while still giving more experienced users analysis patterns they can adapt directly for research. This aligns well with JOSE's emphasis on open educational resources that enable computational learning through authentic practice.

Several design decisions make the Cookbook particularly useful for adoption by other groups. First, each notebook is intended to be self-contained and well-documented, so learners can read, run, and modify a complete workflow without reconstructing missing context from scattered notes. Second, the repository distinguishes tutorials from recipes. Tutorials teach transferable skills, while recipes demonstrate how those skills combine in realistic analysis tasks. Third, the project includes contributor guidance and notebook review conventions that encourage new analyses to be written in a reusable and pedagogically clear style. A caveat is that some recipes use high-resolution model output with datasets

108 of order terabytes, so the underlying data are not always practical to distribute or mirror
109 in full. However, beyond the initial data-loading step, which usually occurs early in each
110 recipe, the analysis workflows are generally applicable to most operational ocean-sea ice
111 model outputs that can be loaded with Xarray (Hoyer & Hamman, 2017).

112 Thus, the structure is valuable beyond the immediate COSIMA context. Many research
113 communities maintain model output on shared infrastructure and face the same challenge:
114 turning expert tacit knowledge into examples that newcomers can adapt. The Cookbook
115 offers a model for how a scientific collaboration can capture that knowledge in version-
116 controlled notebooks, publish it as a living open resource online, and continuously improve
117 it through community contribution.

118 Educational Design and Experience of Use

119 The learning experience is organised as a progression. The documentation points new users
120 first to introductory lessons, including material on loading, slicing, and visualising model
121 output. Learners then move to more advanced tutorials and finally to recipe notebooks
122 that address concrete scientific questions. The categories of “easy” and “advanced” recipes
123 provide a lightweight pedagogical cue about expected complexity and scope, with “regional
124 specialties” specifically covering recipes for regional model configurations (Barnes et al.,
125 2024).

126 The intended mode of use is also explicit. The tutorials and recipes showcase Jupyter-based
127 analysis that can be run directly on Australia’s National Computational Infrastructure,
128 and include guidance on running notebooks through JupyterLab sessions and on accessing
129 shared data holdings. This makes the Cookbook more than a static collection of examples:
130 it is operational documentation for a real analysis environment. At the same time, the
131 notebooks demonstrate broadly transferable workflows for working with labelled geophysical
132 data, so individual lessons can be reused or adapted for specific research questions or for
133 use on other computational platforms.

134 The project also supports community authorship. Contributors are encouraged to submit
135 new notebooks through pull requests, document their methods clearly, and generalise
136 workflows so that they remain useful to others. This is particularly valuable as it is
137 relatively rare for the computer code underlying scientific analyses to be peer reviewed
138 with the same care as the manuscript itself. The COSIMA community provides several
139 examples of the recipe ecosystem being extended into full research projects built on
140 peer-authored software repositories on Github, where analysis workflows, methods, and
141 implementation choices are exposed to community scrutiny and reuse. In that sense, the
142 Cookbook functions both as a learning module for end users and as a framework for
143 teaching reproducible scientific communication through notebook design and peer review.
144 The easy access to data and analysis provided by the COSIMA Cookbook showcases best
145 practices to the broader oceanographic community and has enabled rapid community
146 adoption of the ACCESS ocean and sea-ice model configurations internationally, facilitating
147 to date well over 100 peer-reviewed papers and more than 20 PhD projects to completion
148 to date.

149 Author Order

150 The first two authors contributed equally; authors from the third position onward are
151 listed alphabetically by surname.

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References

- Abernathy, R. P., Busecke, J. J. M., Smith, T. A., Deauna, J. D., Banihirwe, A., Nicholas, T., Fernandes, F., James, B., Dussin, R., Cherian, D. A., Caneill, R., Sinha, A., Uieda, L., Rath, W., Balwada, D., Constantinou, N. C., Ponte, A., Zhou, Y., Uchida, T., & Thielen, J. (2022). *xgcm* (Version v0.8.1). Zenodo. <https://doi.org/10.5281/zenodo.7348619>
- Barnes, A. J., Constantinou, N. C., Gibson, A. H., Kiss, A. E., Chapman, C., Reilly, J., Bhagtani, D., & Yang, L. (2024). regional-mom6: A Python package for automatic generation of regional configurations for the Modular Ocean Model 6. *Journal of Open Source Software*, 9(100), 6857. <https://doi.org/10.21105/joss.06857>
- Constantinou, N. C., Hogg, A. McC., Gibson, A., Steketee, A., Beucher, R., Proft, M., Heerdegen, A., Neme, J., Holmes, R. M., Morrison, A., Fierro-Arcos, D., Kiss, A. E., Oliveira, M., Yung, C., Doddridge, E., Huneke, W., Bhagtani, D., Ong, E. Q. Y., Munroe, J., ... Narayanan, A. (2026). *COSIMA Cookbook: A cookbook of recipes for analysing ocean and sea ice model output*. <https://doi.org/10.5281/zenodo.14353852>
- Dask Development Team. (2016). *Dask: Library for dynamic task scheduling*. <http://dask.pydata.org>
- Delandmeter, P., & Seville, E. van. (2019). The parcels v2.0 lagrangian framework: New field interpolation schemes. *Geoscientific Model Development*, 12(8), 3571–3584. <https://doi.org/10.5194/gmd-12-3571-2019>
- Elson, P., Andrade, E. S. de, Lucas, G., May, R., Hattersley, R., Campbell, E., Comer, R., Dawson, A., Little, B., Raynaud, S., scm72, Snow, A. D., Igolston, Blay, B., Killick, P., Ibdreyer, Peglar, P., Wilson, N., Andrew, ... Kirkham, D. (2024). *SciTools/cartopy: v0.24.1* (Version v0.24.1). Zenodo. <https://doi.org/10.5281/zenodo.13905945>
- Hoyer, S., & Hamman, J. J. (2017). xarray: N-D labeled arrays and datasets in Python. *Journal of Open Research Software*, 5(1), 10. <https://doi.org/10.5334/jors.148>
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>
- Intake developers. (2026). *Intake: A lightweight package for finding, investigating, loading and disseminating data*. <https://intake.readthedocs.io/en/reader/>
- Kiss, A. E., Woodward, M. J., Hollings, T., Mu, M., Fiedler, R., Steven, A. D. L., Lenton, A., Matear, R., Chamberlain, M. A., Oke, P. R., Alves, O., Griffies, S. M., Hill, C., Kovach, A., Marsland, S. J., Fiedler, R., Waters, J., Yang, C.-H., Zhou, X., ... Hogg, A. McC. (2020). ACCESS-OM2 v1.0: A global ocean-sea ice model at three resolutions. *Geoscientific Model Development*, 13(2), 401–442. <https://doi.org/10.5194/gmd-13-401-2020>

198 401-2020

- 199 Kluyver, T., Ragan-Kelley, B., Pérez, F., Granger, B., Bussonnier, M., Frederic, J.,
200 Kelley, K., Hamrick, J., Grout, J., Corlay, S., Ivanov, P., Avila, D., Abdalla, S.,
201 & Willing, C. (2016). Jupyter notebooks—a publishing format for reproducible
202 computational workflows. In F. Loizides & B. Schmidt (Eds.), *Positioning and power*
203 *in academic publishing: Players, agents and agendas* (pp. 87–90). IOS Press. <https://doi.org/10.3233/978-1-61499-649-1-87>
204
- 205 Komiya, T., Brandl, G., Turner, A., Bakosi, J.-F., Shimizukawa, T., Andersen, J. L.,
206 Neuhäuser, D., Finucane, S., Lehmann, R., Mason, J., Kampik, T., Addison, J., Magin,
207 J., Dufresne, J., Waltman, J., Eades, D., Cano Rodríguez, J. L., Ronacher, A., Geier,
208 M., ... Freitag, F. (2025). *sphinx-doc/sphinx: Sphinx 9.1.0* (Version v9.1.0). Zenodo.
209 <https://doi.org/10.5281/zenodo.18109073>
- 210 Rocklin, M. (2015). Dask: Parallel computation with blocked algorithms and task
211 scheduling. In K. Huff & J. Bergstra (Eds.), *Proceedings of the 14th python in science*
212 *conference* (pp. 130–136). <https://doi.org/10.25080/Majora-7b98e3ed-013>
- 213 Zhuang, J., Dussin, R., Bourgault, P., Huard, D., Banihirwe, A., Raynaud, S., Malevich,
214 B., Schupfner, M., Gauthier, C., Filipe, S., Almansí, M., Jüling, A., Scott,
215 R., RondeauG, Rasp, S., Smith, T. J., Meyer, A. G., Mares, B., ... Li, X. (2025).
216 *pangeo-data/xESMF: v0.9.2*. <https://doi.org/10.5281/zenodo.4294774>